

Design of Input and Output Representation of a Neural Network for Predicting Student Dropout Risk in Under-Resourced Schools

(SDG 4 – Quality Education)

Introduction

Student dropout is a major challenge in under-resourced schools due to factors such as poor attendance, financial difficulties, low academic performance, and lack of parental support. Early prediction of students at risk of dropping out enables teachers and school administrators to provide timely interventions. A neural network can analyze multiple student-related factors and accurately predict dropout risk.

Input Representation

The input layer of the neural network consists of features that represent different aspects of a student's academic, social, and economic background. Each feature is converted into numerical values before being fed into the neural network.

Feature	Type	Encoding Method	Justification
Age	Numerical	Min-Max Normalization	Keeps values between 0 and 1, improving training speed.
Gender	Categorical	One-Hot Encoding	Prevents assigning artificial order between categories.
Attendance Percentage	Numerical	Normalization	Attendance strongly influences dropout probability.
Previous Semester Marks	Numerical	Normalization	Academic performance is a major predictor of dropout.
Family Income Level	Categorical	One-Hot Encoding	Socio-economic conditions affect education continuity.
Parents' Education	Categorical	One-Hot Encoding	Educated parents often provide better academic support.
Distance from School	Numerical	Normalization	Long travel distances increase absenteeism.

Feature	Type	Encoding Method	Justification
Internet Availability	Binary	0 = No, 1 = Yes	Indicates access to digital learning resources.
Number of Absent Days	Numerical	Normalization	Frequent absences increase dropout risk.
Participation in Activities	Binary	0 = No, 1 = Yes	Active students are generally more engaged.
Scholarship Received	Binary	0 = No, 1 = Yes	Financial support reduces dropout chances.
Health Issues	Binary	0 = No, 1 = Yes	Chronic health problems may affect attendance.

Feature Encoding Choices

1. Numerical Features

Numerical attributes such as age, attendance percentage, marks, distance from school, and absent days are normalized using Min-Max Normalization. This scales values between 0 and 1, preventing features with larger values from dominating the learning process.

Example:

- Attendance = 85%
- Normalized Value = 0.85

2. Categorical Features

Features such as gender, family income level, and parents' education are encoded using One-Hot Encoding.

Example:

Family Income:

- Low → (1, 0, 0)
- Medium → (0, 1, 0)
- High → (0, 0, 1)

This avoids introducing false numerical relationships between categories.

3. Binary Features

Features such as internet availability, scholarship status, participation in extracurricular activities, and health issues are represented using binary values.

- Yes = 1
- No = 0

Binary encoding is simple, memory-efficient, and suitable for neural network inputs.

Output Representation

The output layer predicts the student's dropout risk.

Binary Classification

Output Value Prediction

0	Student is Not at Risk of Dropping Out
1	Student is at High Risk of Dropping Out

The output neuron uses the **Sigmoid Activation Function**, which produces a probability value between 0 and 1.

Example:

- Output = 0.18 → Low Dropout Risk
- Output = 0.63 → Moderate Dropout Risk
- Output = 0.91 → High Dropout Risk

Schools may define a threshold (e.g., 0.5) to classify students as "At Risk" or "Not At Risk."

Neural Network Structure

Input Layer

- Receives all encoded student features.

Hidden Layer(s)

- Learns relationships among attendance, academic performance, socio-economic background, and other factors.
- Activation Function: ReLU (Rectified Linear Unit).

Output Layer

- Contains one neuron.
 - Activation Function: Sigmoid.
 - Produces the probability of student dropout.
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Advantages of the Selected Encoding

- Normalization improves convergence and training stability.
 - One-Hot Encoding eliminates unintended ordering among categorical values.
 - Binary Encoding is computationally efficient.
 - Multiple educational, financial, and behavioural factors improve prediction accuracy.
 - Probability-based output enables schools to prioritize interventions according to risk level.
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Applications

- Early identification of students at risk of dropping out.
 - Targeted academic counselling.
 - Financial assistance planning.
 - Attendance monitoring.
 - Scholarship allocation.
 - Government education welfare programmes.
 - School performance improvement initiatives.
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Conclusion

Designing suitable input and output representations is essential for building an effective neural network for dropout prediction. Numerical features are normalized, categorical features are one-hot encoded, and binary features are represented using 0 and 1. The output layer uses sigmoid activation to estimate dropout probability. Such a model supports **Sustainable Development Goal 4 (Quality Education)** by helping schools

identify vulnerable students early and implement timely interventions, thereby improving student retention and educational outcomes.